

BACKGROUND

Defining PCO2

Primary cesarean rate in low-risk first-time mothers — the metric driving this review

$$\text{PCO2 RATE} = \frac{\text{Total NTVS (nulliparous, vertex, singleton, term) cesarean deliveries}}{\text{Total NTVS (nulliparous, vertex, singleton, term) all deliveries}} \times 100$$

Nulliparous = first pregnancy · Vertex = head-down · Singleton = one fetus · Term = ≥37 weeks

PREVALENCE

1 in 3

US births are by cesarean delivery.

THE CONCERN



Cesarean rates have climbed without matching gains in maternal or neonatal outcomes — suggesting overuse.

MATERNAL RISK

#1

Cesarean is one of the leading risk factors for maternal mortality and severe morbidity.

OUR ROLE



Weigh short- and long-term tradeoffs & identify safe opportunities to prevent overuse — especially of primary cesareans.

Source: framing adapted from ACOG & CMQCC guidance on safe prevention of primary cesarean delivery.

METHODS

Data Methods

Two analyses on the same NTVS cohort. Both repeated in 2025 to measure change.

01

FOCUS REVIEW

NTVS Cesarean Cases

n = 140

deep-dive on the cesareans only — of 400 NTVS deliveries in 2024

VARIABLES REVIEWED

- Indication for c/s
- Challenges with neuraxial analgesia
- If fetal intolerance: cord pH + APGARs

02

FULL COHORT REVIEW

All NTVS Deliveries

n = 400

every NTVS delivery in 2024 — pathway, demographics, and provider

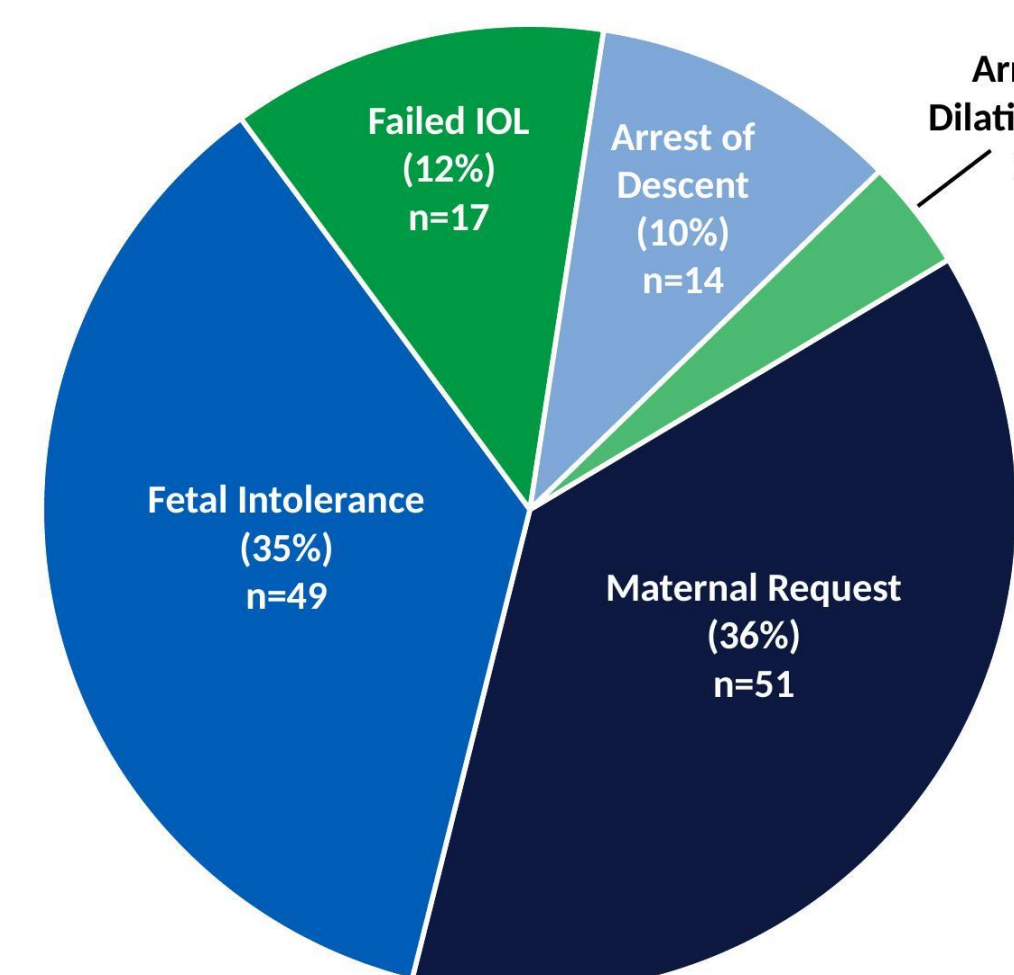
VARIABLES REVIEWED

- Labor onset (active vs. induction)
- Induction type (medical, RR, or term)
- Delivery mode (vaginal vs. cesarean)
- Race, ethnicity, language
- Provider trends

REPEATED FOR 2025 Both analyses applied to the **n = 544** 2025 cohort (Jan 1 – Dec 31, 2025).

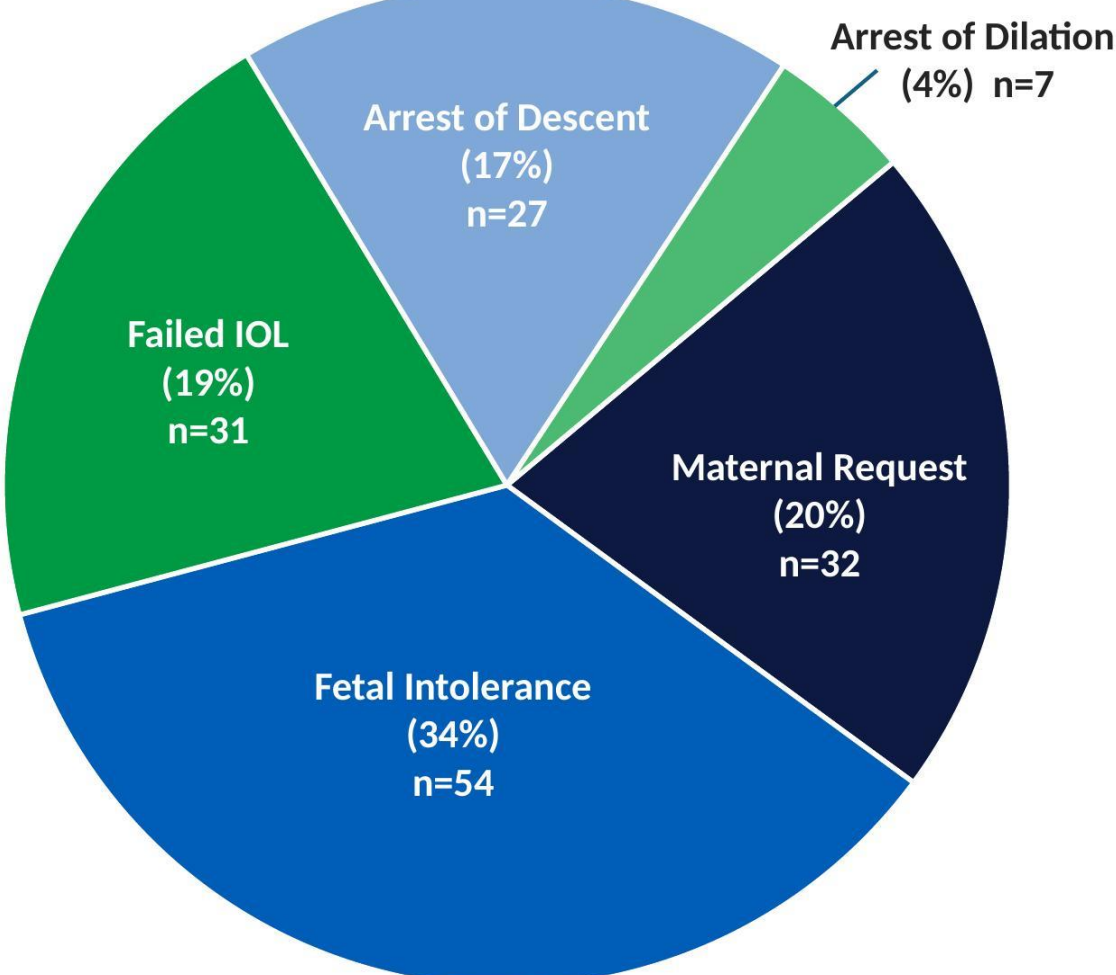
Primary Indications for NTVS

Why cesareans were performed — a side-by-side view of 2024 and 2025



2024 Data N = 140

*33% of indications were changed upon chart review

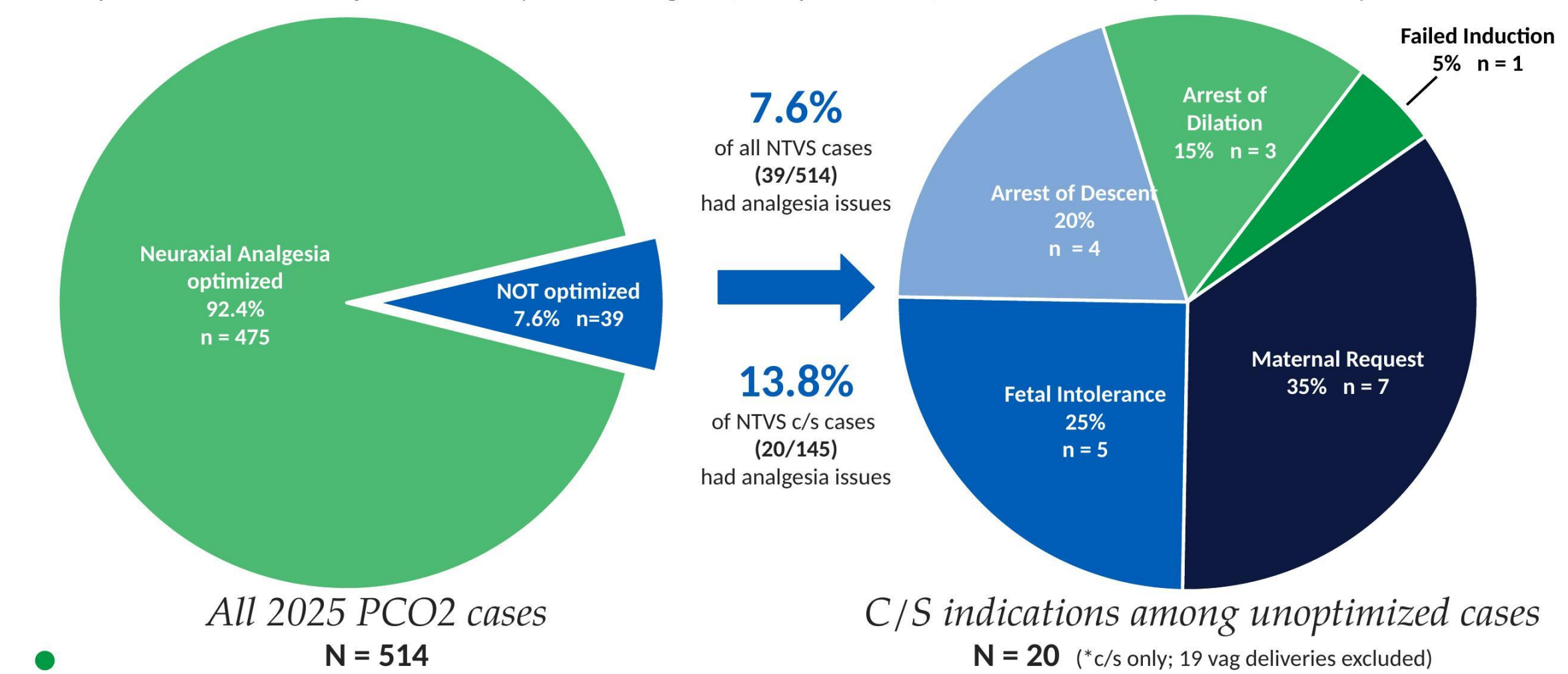


2025 Data N = 160

*26% of indications were changed upon chart review

Pain control improved — but still a driver

Of 2025 NTVS cases, only 7.6% had unoptimized analgesia (down from 19.1%) — but maternal-request remains the top c/s indication



Beyond the Dashboard: A Chart Review-Driven Quality Improvement Initiative to Safely Reduce the Primary Cesarean Rate in a Community Teaching Hospital

Kelly Parker, DO · Sarah Wyand, DO PGY3 · Marissa Coutinho, DO PGY3

Emmanuella Ojukwu, MD PGY2 · Taryn Miracle, MD PGY2 · Carter Waugh, DO PGY1

INDICATION

Fetal Intolerance

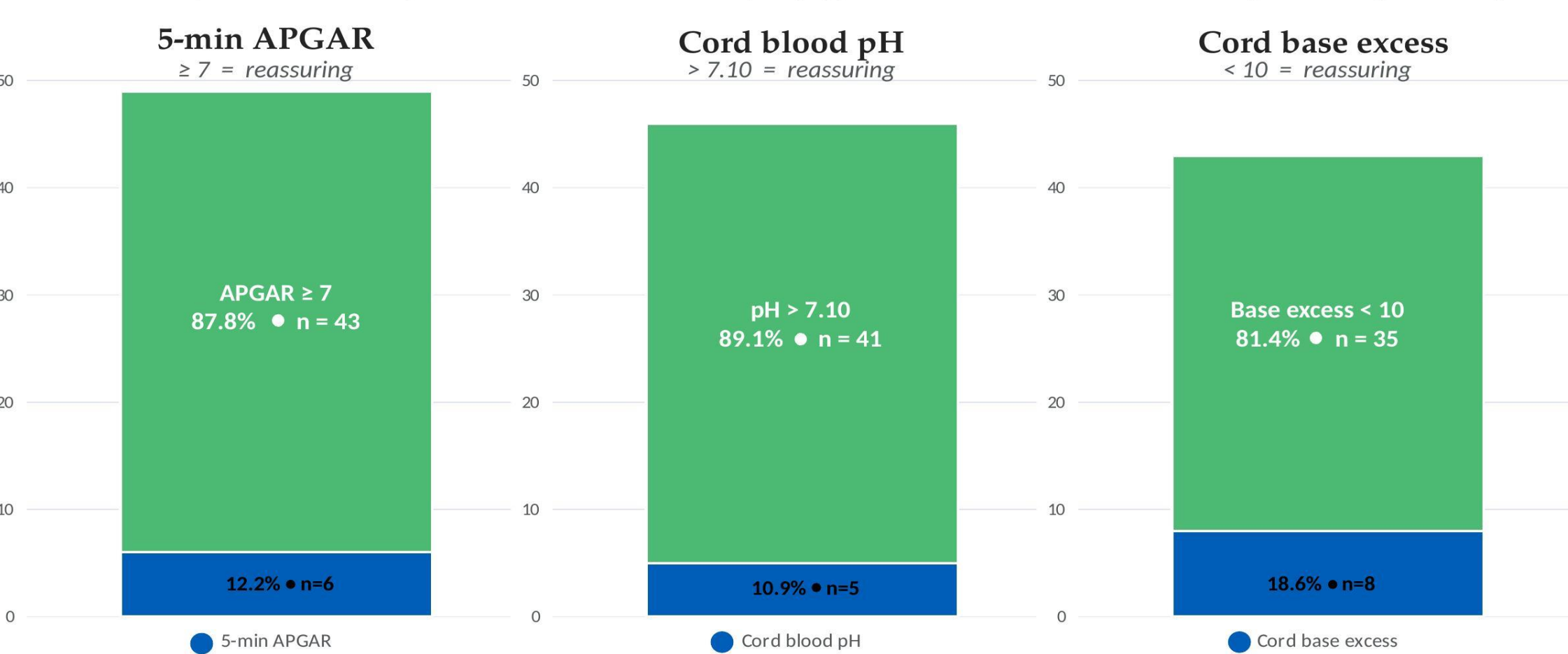
35% → 34% of c/s.

APGAR ≥ 7 rose 87.8% → 96%

2024 FETAL INTOLERANCE · WHAT WE LEARNED

Most fetal-intolerance cesareans delivered babies without compromise

Across three independent measures of newborn distress, the vast majority of fetal-intolerance cases looked clinically reassuring at delivery

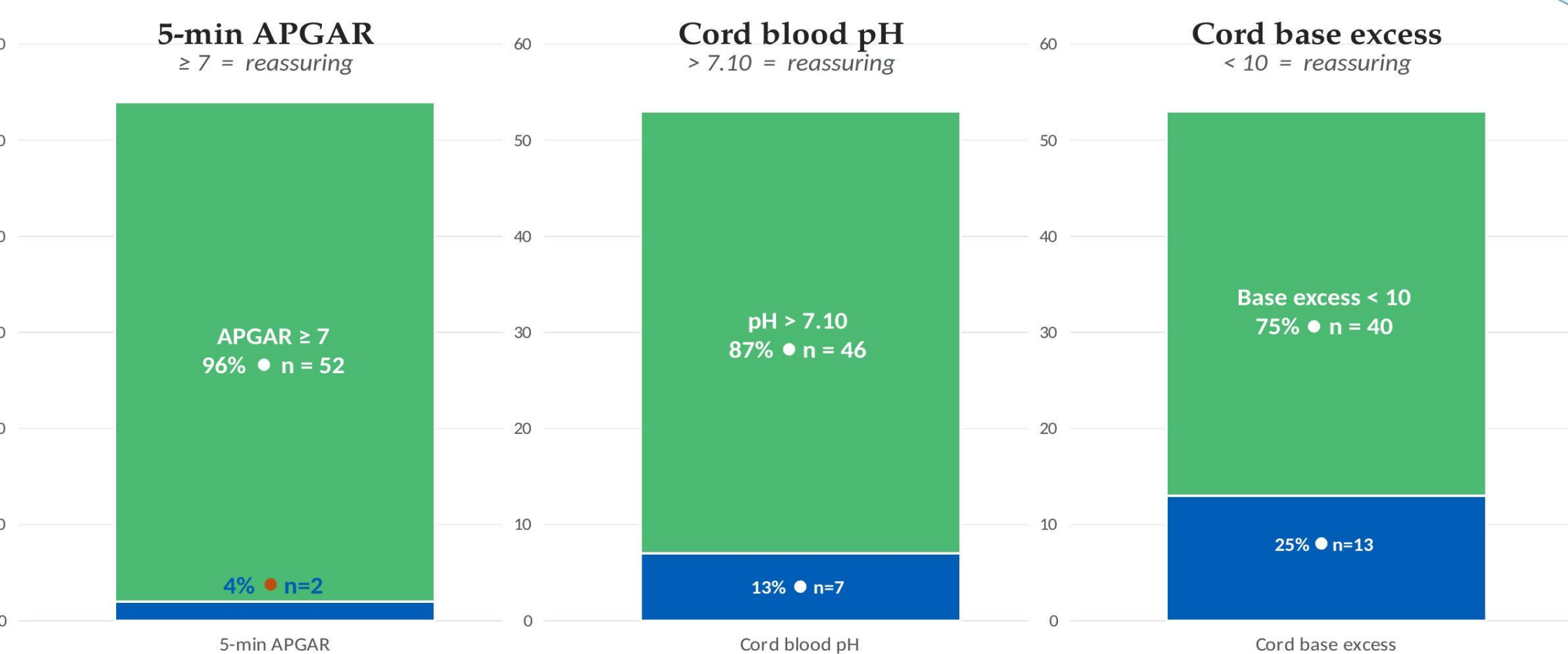


Source: all 49 fetal-intolerance c/s cases in 2024 NTVS cohort. Question: are we calling fetal intolerance too readily?

2025 FETAL INTOLERANCE · WHAT WE LEARNED

We are still overcalling- but becoming more accurate

Most fetal-intolerance c/s still delivered uncompromised babies, but more 2025 cases showed objective biochemical distress



Source: all 54 fetal-intolerance c/s cases in 2025 NTVS cohort. Note: same number of cesarean deliveries for fetal distress, more cord base evidence that distress was present

INDICATION

Failed Induction

12% → 19% of c/s.

Bishop ≤ 3 → 49% c/s rate.

Induction Method and Mode of Delivery

NTVS cohort — January 1, 2024 to December 31, 2025

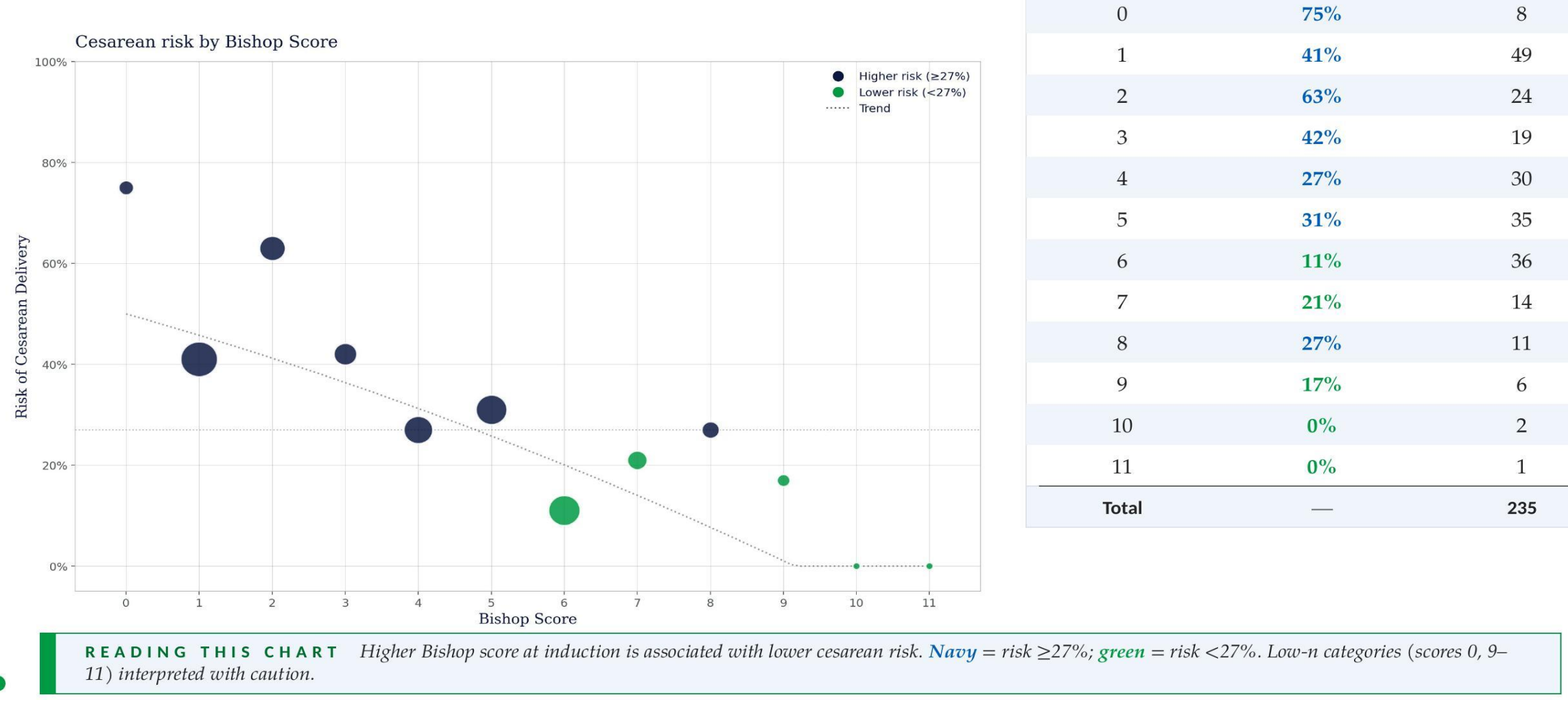
Induction Method	Cesarean Delivery	Vaginal Delivery	p value	Interpretation
Cervidil	8	4	0.0063	Significantly more likely to have cesarean delivery
No Cervidil	98	259		
Misoprostol	111	227	0.2109	No significant difference
No misoprostol	18	54		
Foley balloon	80	151	0.1335	No significant difference
No Foley balloon	49	130		
AROM	87	211	0.1210	No significant difference
No AROM	42	70		
Oxytocin	108	254	0.0678	No significant difference
No oxytocin	21	27		

Fisher exact p value. Red = significantly associated with cesarean delivery; blue = significantly associated with vaginal delivery (none on updated data).

INDUCTION DATA · CERVICAL FAVORABILITY

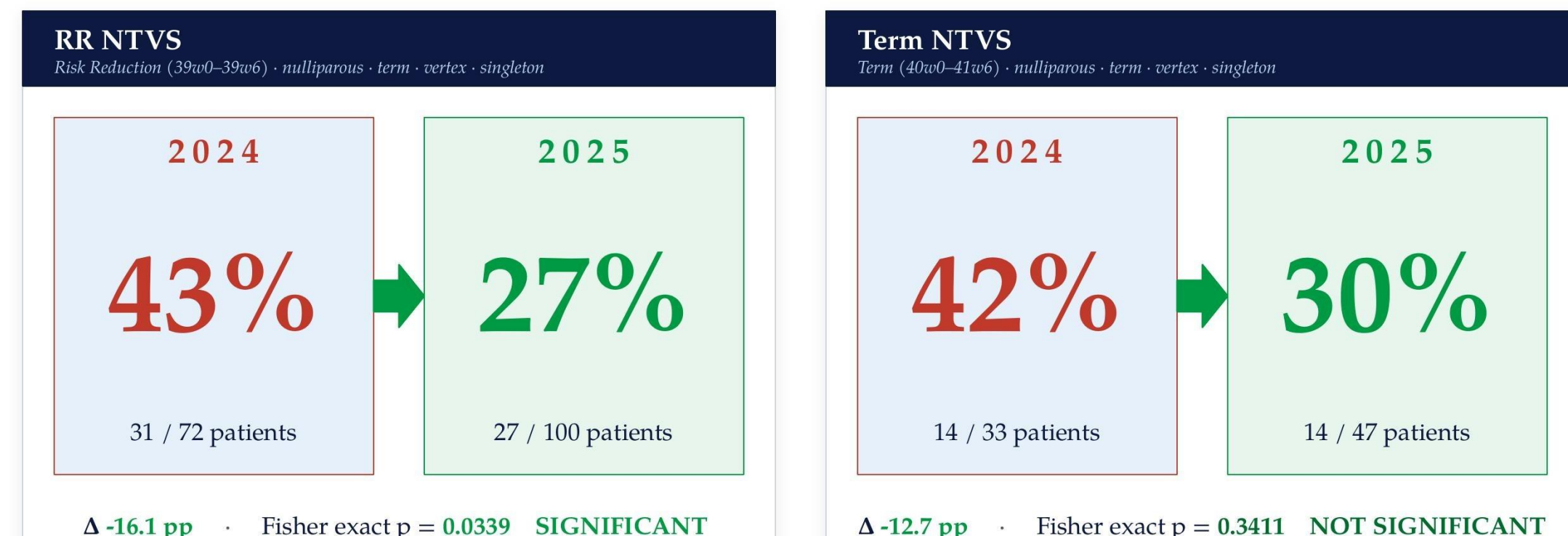
Bishop score predicts cesarean risk

NTVS cohort, 2024–2025 · higher score → lower cesarean risk



NTVS Risk Reducing & Term PCO2 Rate— 2024 vs 2025

Year-over-year change in NTVS risk reducing and term induction cesarean rates, stratified by gestational-age indication



THE NEXT QUESTION · Practice Patterns

Variation across providers — and what changed

CALL GROUPS

All three groups improved

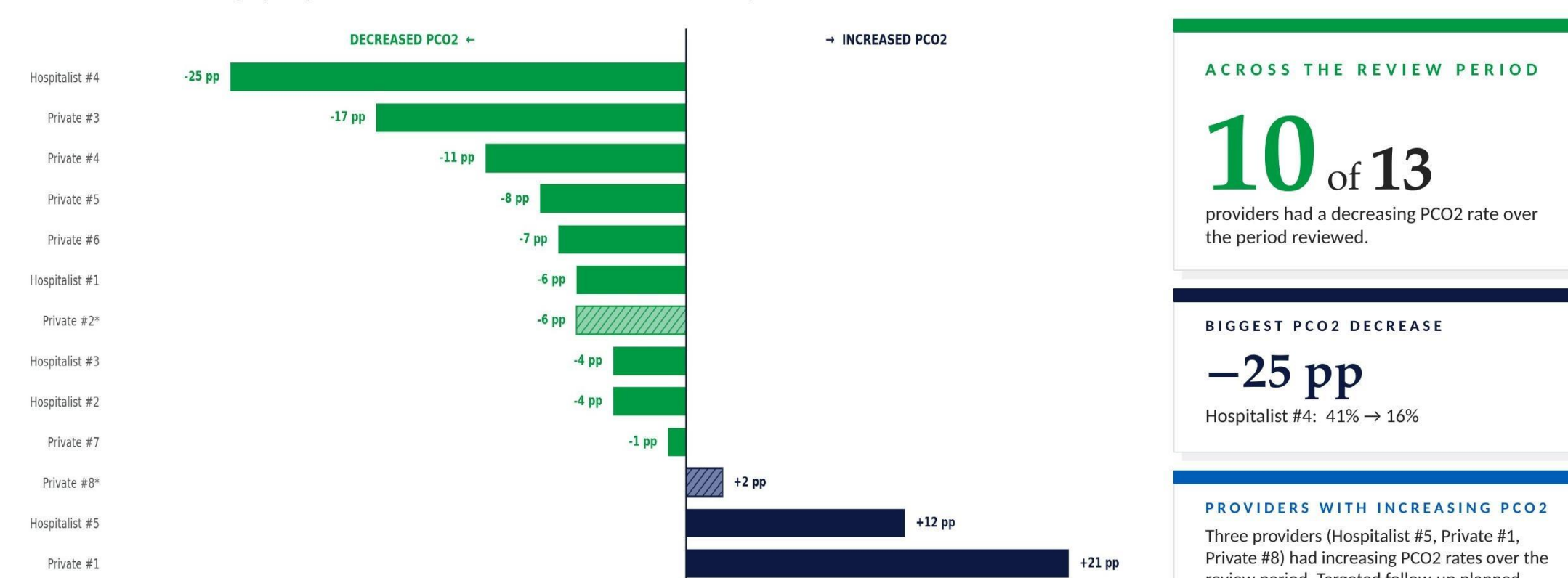
PCO2 rate by call group, 2024 → 2025.



INDIVIDUAL PROVIDER PATTERNS

Most providers decreased their PCO2

PCO2 rate change per provider, 2023 → 2025. Providers anonymized.



THE HARDEST QUESTION · Racial Equity

When the data narrowed — and what came next

SECTION 3 · RACIAL EQUITY

Patients of color compared with white, non-Hispanic patients

A persistent disparity — with parallel improvement across both groups

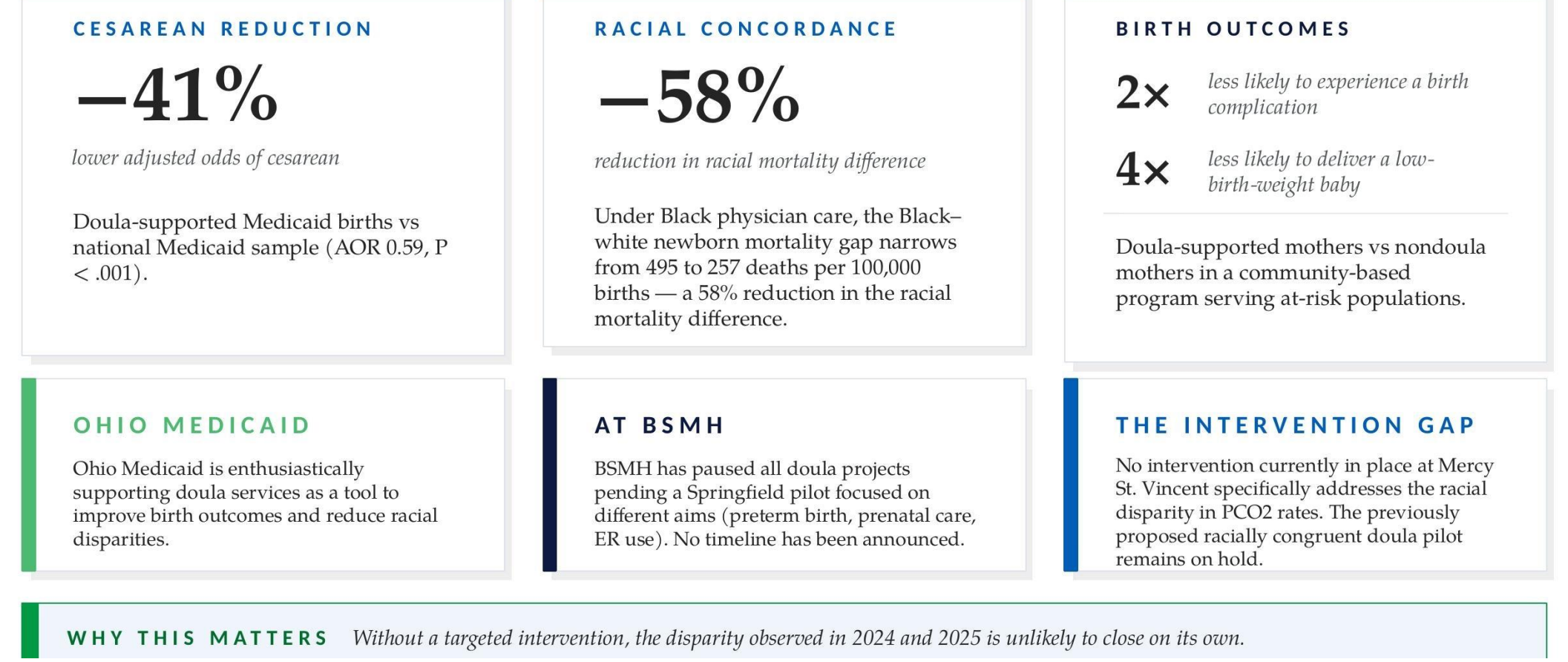


WHAT THIS SHOWS PCO2 decreased in both groups by approximately 5 percentage points; both groups remain above the 24% goal, and the disparity between groups persists. The parallel improvement — rather than disproportionate benefit to white patients — is encouraging but does not close the equity gap. The doula and midwifery interventions that follow are designed to address the disparity directly.

SECTION 3 · RACIAL EQUITY

Racially congruent doula care

An evidence-based intervention specifically targeting the disparity we still see



WHY THIS MATTERS Without a targeted intervention, the disparity observed in 2024 and 2025 is unlikely to close on its own.